

Particle PhysicsNotes

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Lecture 1

1.1 Books

- Peter Renton, *An Introduction to the Physics of Quarks and Leptons*, Cambridge University Press, Cambridge, 1990.
- Francis Helzon, Alan Martin, *Introductory Course in Modern Particle Physics* (Wiley, 1984) (**Old and Standard**)
- Vernon D. Barger and Roger J. N. Phillips, *Collider Physics*, Updated Edition, Addison–Wesley, Reading, Massachusetts, 1997.
- David J. Griffiths, *Introduction to Elementary Particles*, 2nd ed., Wiley-VCH, Weinheim, 2008. (**Easy to Read and follow**)
- David Tong, “Particle Physics” — lecture notes providing a detailed, accessible introduction to particle physics:
<https://www.damtp.cam.ac.uk/user/tong/particle.html> (Cambridge University).:contentReference[oaicite:0]index=0

1.2 Class Timings

- Monday - 11:05 a.m.
- Wednesday - 11:05 a.m.
- Fridays - 09:30 a.m.

Tutor: Utsav Atta

1.3 Marks Distribution

- Attendance + Assignments + Quiz 25%

- Mid-Sem 35%
- End-Sem 40%

1.4 Syllabus

- Experimental Methods + Introduction to Relativistic Kinematics
- Role of Symmetries
- QED + Quark Model
- Deep Inelastic Scattering & QCD¹
- Standard Model
- Beyond Standard Model: Neutrino Physics and Neutrino Oscillations

¹**Deep Inelastic Scattering** is Experimental technique to find internal structure of particle. Find if particle is made of sub-particles. Eg: Rutherford's Scattering Experiment which told us that atom has a nucleus. Calculations done here are not exact (as involves non-perturbative interactions)

Lecture 3

2.1 Language of Particle Physics

- Theoretical/Mathematical $\rightarrow$ QFT $\rightarrow$ Symmetries
- Numerical/Computation
- Data Handling ¹

2.2 Units of Particle Physics

- Quantum Mechanics in SI² $\rightarrow$ lots of $\hbar$ and c
- adapt a unit system where $\hbar = c = 1$
This takes care of not carrying these extra factors

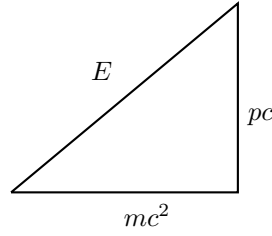
2.2.1 Outcomes of assuming this

- Velocity in special relativity $\beta = \frac{v}{c}$ becomes dimensionless
- Action is dimensionless $\rightarrow$ Energy $\times$ Time
- Mass dimension is same as energy dimension

$$[\text{mass}] = [\text{Energy}] = [\text{momentum}] \xrightarrow{\text{in}} \text{GeV}$$

$$E^2 = p^2 c^2 = m^2 c^4 \rightarrow E^2 = p^2 + m^2$$

¹in LHC large data 1 Trillion bytes per second is generated
²Standard Units

**Fig. 4.1**

- $[\text{Length}] = [\text{Time}] = [\text{Energy}]^{-1}$
 To see this use $c = \frac{\text{Length}}{\text{Time}}$, now since velocity is dimensionless $\implies$ Length and Time have same dimensions.
 Use $\Delta E \Delta t \sim \hbar$, since $\hbar$ is dimensionless Energy will have dimension inverse of time.
- Consider the following table:

Quantity	SI	NU	Unit
Energy	ML^2T^{-2}	1	GeV
Momentum	MLT^{-1}	1	GeV
Time	T	-1	GeV^{-1}
Length	L	-1	GeV^{-1}
Cross Section(σ)	L^2	-2	GeV^{-2}
Action	ML^2T^{-1}	0	dimensionless
Electric Charge	IT	0	dimensionless

Here NU: Natural Units

Cross section is conventionally measured in barns $1 \text{ barn} = 1 \times 10^{-28} m^2$ and need convert accordingly.

- To interpret the result calculated using NU we need to convert it to SI units.
- insert $\hbar$ and c untill the dimensions match back
- $\hbar = 1.05457 \times 10^{-34} \text{ Js}$
 $c = 2.99 \times 10^8 \text{ m/s}$
 $\hbar c = 3.1615 \times 10^{-26} \text{ Jm}$
 $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$
 $1 \text{ MeV} = 1.602 \times 10^{-13} \text{ J}$
 $1 \text{ GeV} = 1.602 \times 10^{-10} \text{ J}$

- Convert J to MeV

$$\begin{aligned}\hbar c &\sim \frac{3.1615 \times 10^{-26}}{1.602 \times 10^{-13}} \text{ MeVm} \\ &\sim 1.97 \times 10^{-13} \\ \hbar c &\approx 197.3 \text{ MeVfm} \quad (\text{fm:femtometer})\end{aligned}$$

Now in natural units $\hbar c = 1$, and hence

$$1 \text{ fm} = \frac{1}{197.3} \text{ MeV}^{-1} = 5.06 \text{ GeV}^{-1} \quad (2.1)$$

- Convert time s to GeV

$$\begin{aligned}\hbar &= 6.58212 \times 10^{-25} \text{ GeVs} \\ 1\text{s} &= \frac{1}{6.58212 \times 10^{-25}} = 1.52 \times 10^{24} \text{ GeV}^{-1}\end{aligned}$$

- Cross Section:

Lecture 4

3.1 Special Relativity

- Laws of physics are same in all inertial reference frames
- the speed of light is constant in vacuum in all inertial frames

$$t' = \gamma(t - \beta x) \quad x' = \gamma(x - \beta t)$$

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} \leftarrow \text{Lorentz Factor}$$

Length Contraction

$$L_0 = \gamma L$$

$$L_0 = x'_B - x'_A \quad L = x_B - x_A$$

★ Proton at LHC has $\gamma \sim 7000$. so the detector appears flattened by a factor of 7000

Time-Dilation

$$\Delta t = \gamma \Delta \tau$$

$$\gamma > 1 \quad \Delta t > \Delta \tau$$

Time dilation increases the lifetime of the particles

Experimental Evidence: Cosmic Rays

Half life of cosmic rays is $\tau_\mu = 2.2\mu\text{s}$. Therefore they can travel the distance $d = c\tau_\mu \sim 660\text{m}$. Hence should decay in atmosphere itself about 10 km high from surface.

But μ^- were observed at surface of earth so lifetime should be more.

3.2 4-Vector Notation

$$x^\mu = (x_0, x_1, x_2, x_3) = (t, x, y, z) = (t, \vec{x})$$

$$x_\mu = (t, -\vec{x})$$

$$x_\mu = \eta_{\mu\nu} x^\nu$$

$$\eta^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

- Invariant Quantity: $a \cdot b = a^\mu b_\mu = a_\mu b^\mu$
- Invariant Interval in space-time

$$\begin{aligned} ds^2 &= dx^\mu dx_\mu \\ &= dt^2 - dx_1^2 - dx_2^2 - dx_3^2 \\ &= dt^2 - d\vec{x} \cdot d\vec{x} \end{aligned}$$

- the sign of ds^2 determines the causal structure

Time-like	Space-like	Light-like
$ds^2 > 0$	$ds^2 < 0$	$ds^2 = 0$
Causal connection is possible	No causal connection possible	Connection through light rays
There exists a frame in which events occur at same place	There exists a frame where events are simultaneous in time	Events lie on light cone

Proper Time

$$\begin{aligned} d\tau^2 &= dt^2 - d\vec{x}^2 \\ \Rightarrow d\tau &= dt \sqrt{1 - \beta^2} \\ dt &= \gamma d\tau \end{aligned}$$

Four-Velocity

$$\begin{aligned} U^\mu &= \frac{dx^\mu}{d\tau} \\ U^\mu &= \frac{dx^\mu}{dt} \frac{dt}{d\tau} = \gamma \frac{dx^\mu}{dt} = \gamma(1, \beta) \end{aligned}$$

U^μ is a normalized four-vector

$$U^\mu U_\mu = 1 \quad (3.1)$$

¹This is particle physics convention also called West-coast convention as people of west USA use this

Four-Momentum

$$p^\mu = mU^\mu = (m\gamma, m\gamma\vec{\beta})$$

$$E = m\gamma = \frac{m}{\sqrt{1-\beta^2}}$$

Mass Shell Condition

$$p^2 = p^\mu p_\mu = m^2 \quad (3.2)$$

$$\implies E^2 - \vec{p}^2 = m^2$$

$$E = \sqrt{|\vec{p}|^2 + m^2}$$

For massless particles: $p^2 = 0 \implies E = |\vec{p}|$ and $\beta = 1$

3.3 Decay Lifetimes

$$t_{\text{lab}} = \gamma\tau = \frac{E}{m}\tau$$

life-time of an unstable particle (τ) is defined in its rest frame

Pion-decay: $\pi^+ \rightarrow \mu^+ + \nu_\mu$ (two body decay)

mass of pion: 140 MeV

$$\tau_{\pi^+} = 2.6 \times 10^{-8}\text{s}$$

Lecture 7

- QFT $\rightarrow$ Calculation tool, provides us with tools to calculate decay rates and scattering cross-section
- This needs an input $\rightarrow$ Lagrangian
- To write lagrangian you need constraints
- Symmetries provide these constraints

Lorentz Group

- $J_i \rightarrow$ Rotation Generators, $K_i \rightarrow$ Boost Generators
- Lorentz Group is $SO(3,1)$
- $[J_i, J_j] = i\epsilon_{ijk}J_k \implies$ forms a closed Algebra, but together J_i and K_i don't form a closed algebra.
- we define $N_i = J_i + iK_i$ and $\bar{N}_i = J_i - iK_i$

4.1 Lie Groups

- Groups with elements as smooth function¹ of some finite set of parameters $\theta_a \in \mathbb{R}$
- $g \in G$ is written as

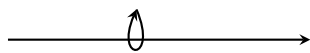
$$g = g(\theta_1, \theta_2, \dots, \theta_n) = e^{i\theta_a T^a} = e^{i\vec{\theta} \cdot \vec{T}} \quad (4.1)$$

where θ_a are continuous parameters

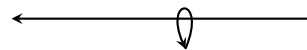
- In $U(1)$ group: e^{ieQ}
where e is charge and Q is generator

¹Continuous and differentiable

- In $SU(2)$ group: $e^{i\theta_a \tau_a}$
 τ_a are Pauli Matrices $\tau_a = \sigma_a/2$
- In $U(N)$ no of parameters = $2N^2$
 $U^\dagger U$ gives N^2 constraints, therefore N^2 elements are there
- In $SU(N)$ has $N^2 - 1$ generators, therefore in 3d 8 generators are there
- from particle physics perspective for each generator there is a mediator
- $U(1)$ describes photons



Spin anti-oriented
 along momentum
Left handed



Spin oriented
 along momentum
Right handed

- Parity group $\mathbb{Z}_2$

$\cdot$	e	a
e	e	a
a	a	e

- $\mathbb{Z}_n$ roots of $z^n - 1$

Lecture 8

- Matter
 - Leptons
 - Quarks
- Force
 - QED
 - QCD
 - Weak
- To classify particles we need quarks
- We need Isospin ($SU(2)$)
- we need flavor ($SU(3)$)
- Interchange, decays, scattering dictated by conservation of symmetries (I_3 , strangeness)
- other symmetries
 - discrete
 - charge
 - parity

Benefits of Group Theory

- We write transformations (symmetry) whose origins are in group theory
- Associated internal symmetries (Noether currents)
- Structure of Lagrangian in gauge theories $\implies$ SM Lagrangian
- Symmetries such as CPT associated with discrete groups

- IN SM the relevant groups are $U(1)$: QED & Hypercharge (y)
 $SU(2)$: Electroweak symmetry and spin
 $SU(3)$: QCD and Hadron states
- SM Symmetry Group¹: $SU(2)_L \times U(1)_Y \times SU(3)_C$
- We know about space time symmetries
 - translation
 - Rotation and Boost²
 - parity in (x,t) reflection
- $[J_i, J_j] = i\epsilon_{ijk}L_k$ $SU(2)$ algebra
- $[K_i, K_j] = i\epsilon_{ijk}L_k$
- The group is $SO(1,3) = SU(2) \oplus SU(2)$ with $\det \pm 1$
- $N_i^\pm = J_i \pm iK_i$

$$[N_i^\pm, N_j^\pm] = i\epsilon_{ijk}N_k^\pm$$

- Translation $x \rightarrow x' = x - \Delta x$; $t \rightarrow t' = t - \Delta t$
- Rotation about z -axis

$$x \rightarrow x' = x \cos \theta - y \sin \theta$$

$$y \rightarrow y' = x \sin \theta + y \cos \theta$$

- In QM $x' = f(x)$, Inverse it $x = f^{-1}(x')$
 for Example: $x' = e^x$ $\psi(x) = x \implies x = \log x$ $\psi'(x') = \log x'$
-

$$\begin{aligned}\psi &= \sum c_i \phi_i \\ \psi' &= U \sum c_i \phi'_i = \sum c_i U \phi'_i \\ \langle \psi' | \psi' \rangle &= \langle \psi | \psi \rangle \implies U^\dagger U = \mathbb{1}\end{aligned}$$

- we want to deal with unitary operators

$$\begin{aligned}\langle \psi | A | \psi \rangle &= \langle \psi' | A' | \psi \rangle \\ &= \langle U \psi | A' | U \psi \rangle \\ &= \langle \psi | U^\dagger A' U | \psi \rangle\end{aligned}$$

$$A' = U A U^\dagger \tag{5.1}$$

¹Means that the Lagrangian should respect this symmetry

²translation, rotation and boost together form *Poincare Group*

- let A be invariant under this transformation then

$$[U, A]$$

- Now in context of fields

$$U = e^{iaX} \tag{5.2}$$

a is real and x is the generator of the transformation